

OSHADHA SAMARAKOON

Undergraduate – Faculty of Engineering, University of Peradeniya

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RESEARCH INTEREST

I study how visual representations form inside modern architectures, first by benchmarking visual state-space models against CNNs and Vision Transformers on remote-sensing segmentation, change detection, and solar-irradiance forecasting, and now moving toward a more mechanistic question: what these encoders actually learn, and how that compares to biological vision. I work within the Multidisciplinary AI Research Centre (MARC) at the University of Peradeniya, leading architecture design and experiments of ongoing project **SolarMamba** (irradiance forecasting) and working on project **ORBIT-Mamba** (change detection).

HIGHLIGHTS

- **Published researcher:** Three papers (IGARSS 2026; two published at MERCon 2026) spanning remote-sensing change detection and multimodal solar-irradiance forecasting.
- **Research group lead:** Leads a small research group at MARC developing state-space (Mamba) architectures for Solar Irradiance prediction; currently building **ORBIT-Mamba** (targeting IEEE TGRS) and **SolarMamba** (targeting Applied Energy).
- **Competitive programmer:** ICPC Sri Lanka Regional Finalist; IEEE Xtreme 18.0 Sri Lanka Top 100; multiple national contest podium and finalist finishes.
- **Open research presence:** Active GitHub and Hugging Face profiles publishing benchmark code, trained checkpoints, and a personal research portfolio/blog on state-space models and signal processing.

PUBLICATIONS

Peer-Reviewed Conference Papers

[C.1] N. Wasalathilaka, D. Perera, **O. Samarakoon**, B. Wijenayake, R. Godaliyadda, V. Herath, and P. Ekanayake
A Controlled Benchmark of Visual State-Space Backbones with Domain-Shift and Boundary Analysis for Remote-Sensing Segmentation

IEEE IGARSS, 2026. Accepted.

[\[paper\]](#) [\[code\]](#) [\[HF\]](#)

Contribution: main ideation, literature survey, implementation, and writing.

[C.2] D. Perera, T. Firdous, **O. Samarakoon**, R. Godaliyadda, P. Ekanayake, and V. Herath
MambaRefine-CD: MambaVision with Region–Boundary Temporal Refinement

MERCon, 2026. Published. [\[DOI\]](#)

[\[paper\]](#) [\[code\]](#) [\[HF\]](#)

Contribution: pipeline design, experiment planning, and implementation.

[C.3] **O. Samarakoon**, D. Herath, I. Ranmandala, D. Herath, R. Godaliyadda, P. Ekanayake, and V. Herath
A Controlled Visual-Backbone Benchmark for Multimodal Short-Term Solar Irradiance Forecasting

MERCon, 2026. Published. [\[DOI\]](#)

[\[paper\]](#) [\[code\]](#) [\[project\]](#) [\[HF\]](#)

Contribution: Led the research end-to-end; co-authors supervised and refined the implementation.

Manuscripts in Preparation

[P.1] **O. Samarakoon**, D. Herath, et al.

SolarMamba: Adaptive Fusion of Sky-Camera Imagery and Temporal Weather Data for Short-Term Solar Irradiance Nowcasting

Target: *Applied Energy*.

[\[code\]](#)

In preparation.

[P.2] Dineth Perera, **O. Samarakoon**, Thaarig Firdous, R. Godaliyadda, et al.
ORBIT-Mamba: A State-Space Architecture for Remote-Sensing Change Detection
Target: *IEEE TGRS*.
In preparation.

EDUCATION

University of Peradeniya — B.Sc. Engineering (Honours), Electrical & Electronic Engineering Mar. 2023 – Jun. 2028 (expected)

Faculty of Engineering, Department of Electrical and Electronic Engineering. *CGPA: 3.504 / 4.00*.

G.C.E. Advanced Level Examination — Physical Science Stream Mar. 2022

3 Distinction (A) passes: Combined Mathematics, Physics, and Chemistry. Ranked **5th in Kandy District** and **98th islandwide** (Z-score 2.5756); best-ever result in school history.

G.C.E. Ordinary Level Examination Dec. 2018

9 Distinction (A) passes.

RESEARCH EXPERIENCE

Undergraduate Researcher — Multidisciplinary AI Research Centre (MARC), University of Peradeniya Aug. 2025 – Present

Supervisor: Prof. Roshan Godaliyadda.

- Conduct research on visual state-space models, remote-sensing segmentation and change detection, and multimodal solar-irradiance forecasting.
- Lead architecture design, training, and experimental work for the **ORBIT-Mamba** and **SolarMamba** projects within the group.
- Contributed to three accepted conference papers through ideation, literature review, experiment planning, implementation, and manuscript writing.

Research Assistant — National Institute of Fundamental Studies (NIFS) Oct. 2022 – Mar. 2023

Supervisor: Prof. G. R. A. Kumara.

- Investigated supercapacitor performance using activated carbon, gaining practical experience in device fabrication and materials research.
- Fabricated devices under laboratory safety procedures, analyzed experimental data in Origin, and prepared literature reviews and references.

RESEARCH PROJECTS

• **SolarMamba** — *Multimodal solar-irradiance nowcasting* Oct. 2025 – Present

Originated the research idea and led architecture design, end-to-end training, and experiment planning for adaptive fusion of sky-camera imagery and temporal weather data. First-listed author, equal contribution with D. Herath. [\[code\]](#)

• **ORBIT-Mamba** — *Remote-sensing change detection* Mar. 2026 – Present

Designing a state-space architecture for change detection within MARC; responsibilities span architecture, experiment design, literature review, and manuscript writing.

• **Image Encoders** — *Controlled study of modern vision backbones* Ongoing

Custom implementations and controlled experiments across CNN, Vision Transformer, and Mamba/SSM encoders to compare feature extraction, architectural behavior, and computational trade-offs. [\[project\]](#)

• **BraTSMamba** — *3D brain-tumor segmentation* Dec. 2025 – Mar. 2026

Led the research team for the BioFusion Biomedical Deep Learning Hackathon; shared equal contributions with D. Perera in ideation, architecture design, training, and manuscript writing. Built sub-region segmentation on BraTS 2021 (Task 1) using a linear-complexity Mamba backbone for full-resolution multimodal MRI. [\[code\]](#)

- **PatchFlow-PdM** — *Synthetic fault-data generation* Jan. – May 2026
Led Team Astra in the IEEE IES Generative AI Challenge 2026; directed research formulation, designed the architecture, trained models, and prepared the methodology section and figures. Advanced through Milestone 3; manuscript recommended for IRAI 2026.
- **CV Solar Irradiance** — *All-sky-image irradiance forecasting pipeline* Research study
MARC/DLR-linked pipeline combining OCR metadata extraction, PSA irradiance synchronization, classical red/blue-ratio and deep cloud segmentation, and short-horizon GHI/DNI/DHI forecasting. [\[code\]](#) [\[project\]](#)

ENGINEERING & SYSTEMS PROJECTS

- **Omni-Wheel Autonomous Robot** — *ROS + Jetson, LiDAR/Kalman fusion, SLAM, PID; exhibited at EngEx 2025* Completed
[\[project\]](#)
- **Symon Says** — *Sound-controlled memory game; MATLAB–ESP32 real-time audio classification, sub-100 ms inference; exhibited at EngEx 2025* Completed
[\[code\]](#) [\[site\]](#)
- **ConTXT BOX** — *Local-first MCP context layer for coding agents; lazy workspace indexing, retrieval, on-demand extraction* Active
[\[code\]](#)
- **Server Sentinel C** — *C99/GTK4 data-center monitoring simulator with state-machine logic, alerting, and portable Windows distribution* Completed
[\[code\]](#)
- **Li-Fi File Sharing** — *Visible-light communication file-transfer system* Completed
[\[code\]](#)

AWARDS & HONORS

- **ICPC Sri Lanka Regional Finals 2025/26** — qualified for regional finals; Top 15 team finish.
- **PreCoders V11, 2024** — 1st place (team), University of Peradeniya; Coders V11: 42nd place.
- **IEEE Xtreme 18.0** — Global Top 900; Sri Lanka Top 100.
- **CodeArena** — 3rd place in selection; finalist (Top 20 of 150 teams).
- **MoraXtreme 2025** — finalist among 150 teams; **Jaffna Coders V4**: Top 15; **Haxtreme**: 8th place.
- **PreCoders V10** — Top 25, selected for Coders V10 Final; **CodeRally**: 2nd place (Beginner Tier selection), finalist.

TECHNICAL SKILLS

- **Programming**: Python, C, C++, MATLAB, $\LaTeX$, Altium.
- **Machine Learning**: PyTorch, JAX, Hugging Face; Mamba/SSM, Vision Transformers, CNNs, conditional flow matching.
- **Robotics & Embedded**: ROS, Arduino, ESP32, NVIDIA Jetson, OpenCV.
- **Research**: experimental design, literature review, manuscript writing, scientific figure preparation, Origin.
- **Systems**: Git, GitHub, Linux.

INDEPENDENT STUDY & TECHNICAL WRITING

- **Distance Geometry in ML** — distance-geometry techniques applied to learned representations. 
- **Information Theory** — Shannon theory, Huffman coding, information bottleneck. 
- **Nonlinear & Multivariable Systems** — Lyapunov stability, bifurcations, feedback linearization. 
- **“The Eternal Harmonic: A Tribute to Joseph Fourier”** — essay tracing signal processing from 19th-century heat theory to Transformer positional encodings. [\[writing\]](#)
- **“From Apollo to AI: A Deep Look into State Space Models”** — essay on the lineage from Kalman filtering to the Mamba architecture. [\[writing\]](#)